Proceeding Series of the Brazilian Society of Computational and Applied Mathematics

Assessing Vulnerability through Graph Centrality and Efficiency Analysis: A Geospatial Analysis of the Tamanduateí River Basin

Emmanuel I. Babajide[1], Giovanni G. Soares[2]
Applied Computing, National Institute for Space Research (INPE), São José dos Campos, SP, Brazil

Rogério G. Negri[3]
São Paulo State University (UNESP), São José dos Campos, São Paulo, Brazil

Elton V. Escobar-Silva[4], Leonardo B. L. Santos[5]
National Center for Monitoring and Early Warning of Natural Disasters (Cemaden), São José dos Campos, São Paulo, Brazil

Abstract. This study analyzes the vulnerability of an urban street network under flood-related disruptions using a graph-theoretical framework applied to the Tamanduateí River Watershed, São Paulo, Brazil. Intersections are modeled as nodes and road segments as edges, with betweenness centrality quantifying node importance and vulnerability defined as the relative reduction in global network efficiency after node removal. Historical flood records (2018–2025) were spatially matched to network nodes, and their distributions compared against randomly sampled non-flood nodes using KDE, ECDF, and non-parametric statistical tests. Flood-affected nodes exhibited significantly higher betweenness centrality and vulnerability ($p < 10^{-6}$), with median values 156% and 49% greater than non-flood nodes respectively, corresponding to a 0.24% versus 0.16% relative reduction in global network efficiency upon removal. These findings confirm that flood events are disproportionately associated with structurally critical intersections, amplifying network disruption risk.

Keywords. Graph Theory; Urban Transportation Networks; Betweenness Centrality; Global Efficiency; Network Vulnerability; Flood Susceptibility.

1 Introduction

Urban transportation systems provide the basic infrastructure needed to support economic activity, social mobility and access to essential services within modern cities [1]. However, these systems are increasingly disrupted due to weather events such as severe flooding [11]. Flood events may result in the temporary or permanent interruption of road segments, significantly reducing network connectivity and limiting accessibility across the urban space [2, 3]. As climate-related hazards become more frequent and intense, understanding the structural vulnerability of transportation networks has become an important topic in infrastructure planning and disaster risk management [10].

[1]babajideemma@gmail.com
[2]giovanniguarnierisoares@gmail.com
[3]rogerio.negri@unesp.br
[4]eltescobar@gmail.com
[5]santoslbl@gmail.com

From a mathematical perspective, transportation systems can be modeled as complex networks in which intersections are represented as nodes and road segments as edges [12]. This abstraction allows the application of graph theory and network science to analyze structural properties, and graph-based approaches have been successfully applied to examine structural organization, mobility patterns, and system vulnerability in urban transportation [3, 6]. Among centrality measures, betweenness centrality is particularly relevant as it identifies elements lying on a large number of shortest paths, indicating potential bottlenecks or critical corridors [5]. Complementarily, global efficiency measures how efficiently movement can occur across the network through shortest paths [9], and combining these measures provides a robust framework for assessing transportation network vulnerability and resilience [8, 13].

The aim of this work is to assess the vulnerability of an urban road network using graph-theoretical metrics and efficiency-based analysis. The objectives are: (i) to represent the urban road network as a spatial graph using geospatial data; (ii) to compute centrality measures to identify structurally important nodes and links; (iii) to conduct a comparative analysis of betweenness centrality and vulnerability distributions between flood-affected and randomly selected non-flood nodes using statistical tests; (iv) to evaluate global network efficiency under disruption scenarios; and (v) to analyze how the removal of critical components affects overall connectivity and network performance.

2 Methodology

2.1 Study Area and Data

The study focuses on the Tamanduateí River Watershed (TRW) located in the southeastern region of the São Paulo Metropolitan Area, Brazil. The basin covers approximately 330 km^2 and drains into the Tietê River. Historically, the river corridor supported industrial development, but it has also been associated with recurring flood events [7].

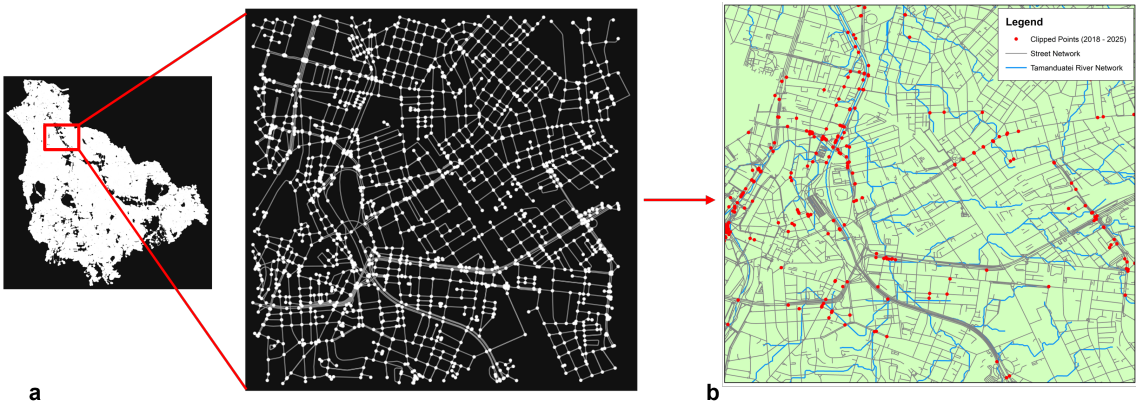


Figure 1: Study Area (a) the street network graph and (b) the historical flood points.

The street network was extracted from OpenStreetMap using the OSMnx Python library [4]. The analysis considered drivable streets of 4 km×4 km around the coordinate (23.55 °S; 46.62 °W) to define the study network (Figure 1). The street network was modeled as a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where nodes $\mathcal{V} = \{v_1, \ldots, v_n\}$ represent intersections and edges $\mathcal{E} = \{e_{ij} : i, j = 1, \ldots, n\}$ in the

directed graph represent street segments, with each edge weighted by its actual road segment length (in metres) as extracted from OpenStreetMap data.

Historical flood incident data from 2018–2025 were provided as spatial point datasets. Each flood point was associated with the nearest network node within a 100 m radius, generating the set of flood-affected nodes. This threshold was selected to account for geolocation uncertainty in historical reports, spatial offsets between reported flood coordinates and mapped intersections, and local street-network spacing.

A control group of random non-flood nodes was sampled from nodes located outside the 100 m matching radius. The sample size was matched to the number of flood-affected nodes to preserve balanced comparison and reduce sampling bias.

2.2 Network Metrics

This section details the methodologies employed for calculating key network metrics: betweenness centrality, global efficiency, and node vulnerability, which are crucial for understanding the urban network's structure and resilience.

Betweenness Centrality

Betweenness centrality measures the proportion of shortest paths passing through a node v [5]:

$$BC(v) = \sum_{s,t \neq v} \frac{\sigma_{st}(v)}{\sigma_{st}} \tag{1}$$

where σ_{st} is the total number of shortest paths between nodes s and t, and $\sigma_{st}(v)$ is the number of those paths that pass through node v.

In this study, shortest paths were computed using Dijkstra's algorithm with road-segment length (meters) as edge weight, based on OpenStreetMap length attributes. Therefore, nodes with high betweenness centrality are those frequently traversed along minimum-distance travel routes.

Global Efficiency

Following Latora and Marchiori [9], the global efficiency of a graph $\mathcal{G}$ is defined as:

$$E(\mathcal{G}) = \frac{1}{N(N-1)} \sum_{s \neq t} \frac{1}{d_{st}} \tag{2}$$

where N is the total number of nodes and d_{st} is the weighted shortest path distance between nodes s and t calculated as the minimum cumulative road length between nodes. Global efficiency is the average of the inverse of the distance for every pair of nodes.

For disconnected node pairs, where no path exists, distance is considered infinite and their the efficiency contribution for these pairs becomes zero ($1/d_{st}$=0). This approach ensures that the node vulnerability metric (V_v) accurately captures the significant loss of connectivity and accessibility across the urban space when critical intersections are compromised

Node Vulnerability

Node vulnerability measures the relative loss of global efficiency in the network after a specific node v is removed [5]:

$$V_v = \frac{E(\mathcal{G}) - E(\mathcal{G} - v)}{E(\mathcal{G})} \tag{3}$$

Higher values of V_v indicate nodes whose failure or removal strongly degrades the overall network performance.

2.3 Statistical Analysis

To compare the differences in node group distributions (flood-affected versus non-flood), a two-step analysis was performed. First, kernel density estimation (KDE) visualized the probability density functions of centrality and vulnerability measures, alongside empirical cumulative distribution functions (ECDFs) to track cumulative shifts. Then, two nonparametric tests—the Kolmogorov–Smirnov (KS) test and the Mann–Whitney U test—were used to determine if there are significant distributional differences between the groups, without assuming normality. To complement the significance tests and provide a substantive interpretation of the magnitude of group differences, the median values of betweenness centrality and vulnerability are reported for each group, expressed as percentage differences and in terms of relative network efficiency loss.

In the context of urban transportation systems modeled as directed graphs, nodes and edges possess inherent structural and spatial dependencies due to their topological connectivity and the functional relationships within the network. This limitation is common in complex network analysis, and these tests are used here to provide a statistically significant quantitative comparison of how environmental hazards intersect with network criticality.

3 Results and Discussion

3.1 Centrality Distribution Analysis

The distribution of betweenness centrality across the network is highly skewed (Figure 2), indicating that a small number of intersections serve as key connectors. Spatial visualization reveals that high-centrality nodes are concentrated along major arterial routes, especially near Tamanduateí Avenue.

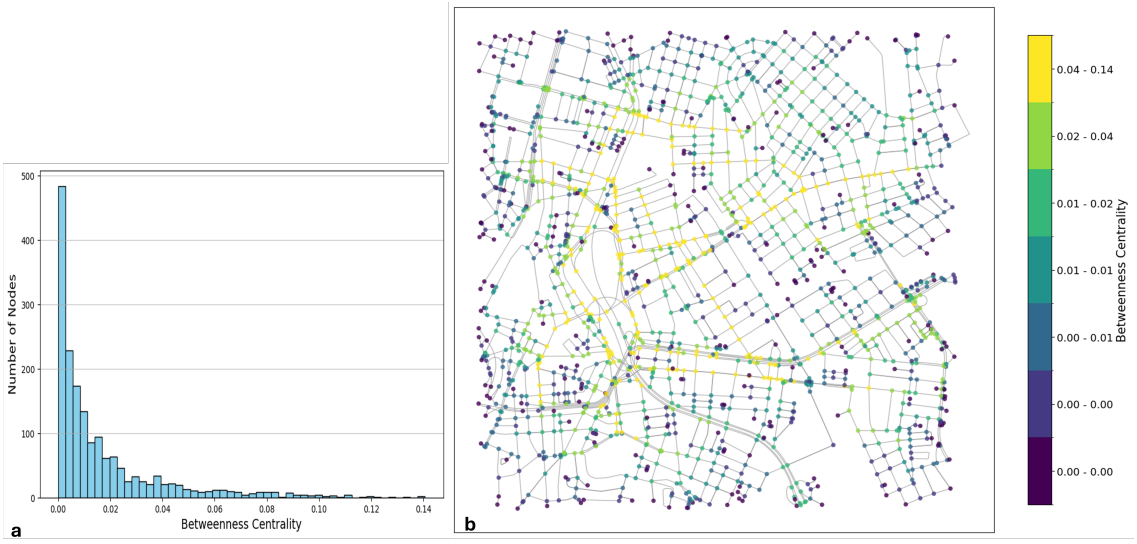


Figure 2: Centrality Distribution Analysis for all Nodes (a) statistical distribution and its (b) geospatial distribution.

Comparison of KDE plots reveals a distinct rightward shift for the flood-affected group compared to the general network and the random control group (Figure 3). This plot strongly suggests that areas experiencing flooding are not just random spots on the map, but they often coincide with critical junctions and segments that are vital for network flow. The fact that flood-affected nodes exhibit higher centrality even when compared to other nodes in the same sample area highlights a systemic vulnerability where environmental hazards intersect with network criticality.

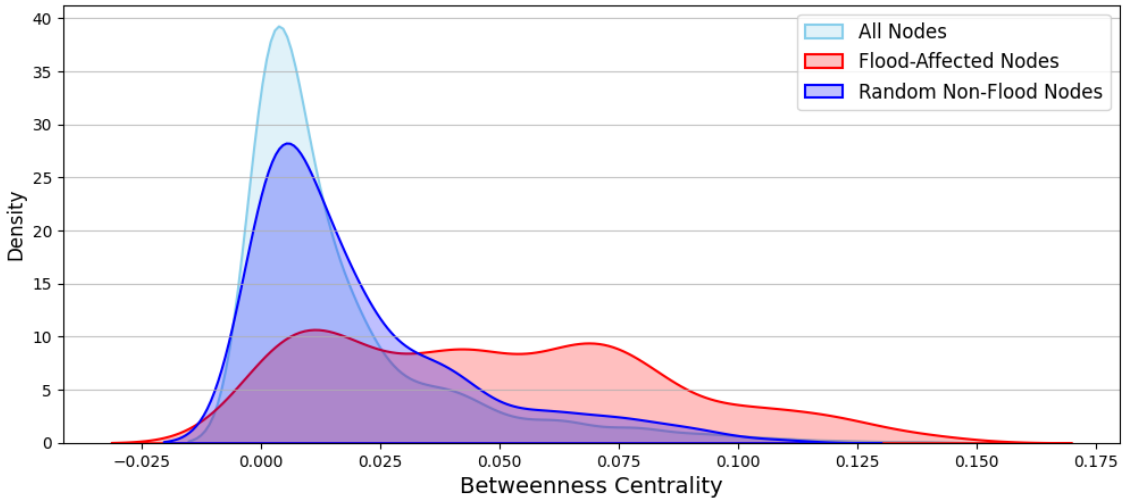


Figure 3: KDE plot comparing all node categories.

3.2 Vulnerability Distribution Analysis

While the majority of nodes in the network exhibit low vulnerability, the flood-affected nodes shows a higher density in the high-vulnerability tail (Figure 4a). ECDF plots (Figures 4b) confirm that for any given percentile, flood-affected nodes consistently demonstrate higher centrality and vulnerability values. This confirms that their removal causes a disproportionate loss in global efficiency compared to non-flood nodes.

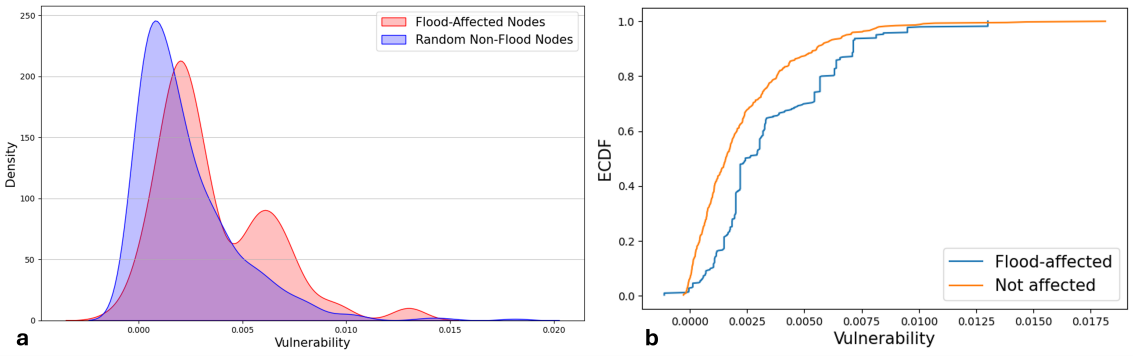


Figure 4: KDE and ECDF Plot Comparing Node Vulnerabilities

3.3 Statistical Significance Testing

The Kolmogorov–Smirnov and Mann–Whitney U tests indicated statistically significant differences in betweenness centrality between flood-affected and non-flood nodes ($p < 0.001$) as seen in Table 1. This indicates that the observed differences in betweenness centrality distributions between flood-affected nodes and random non-flood nodes are highly statistically significant.

Similarly, vulnerability values were significantly higher among flood-affected nodes ($p < 0.01$), indicating that the observed differences in vulnerability distributions between flood-affected nodes and random non-flood nodes are highly statistically significant.

Table 1: Statistical validation and effect size measures for metric distributions (Flood-Affected vs. Random Non-Flood Nodes).

Metric	**Med. (Flood)**	**Med. (Control)**	**Statistical Test**	***p*-value**	**Effect Size**
Betweenness Centrality	0.0276	0.0108	Kolmogorov–Smirnov	9.037×10^{-37}	156% higher
			Mann–Whitney U	1.706×10^{-37}	
Node Vulnerability	0.0024	0.0016	Kolmogorov–Smirnov	1.79×10^{-18}	49% greater
			Mann–Whitney U	7.40×10^{-17}	

In addition to being statistically significant, the magnitude of difference between groups indicates that these effects are substantial. The median betweenness centrality of flooded nodes was 0.0276, which is about a 156% increase over the non-flooded node's median of 0.0108. With respect to vulnerability, the flooded nodes had a median value of 0.0024 and the non-flooded nodes had a median value of 0.0016, corresponding to a median network efficiency decrease upon node removal of 0.24% (flooded nodes) and 0.16% (non-flooded nodes), respectively. Therefore, removing a flooded node from the global network would result in an approximate relative loss of global network efficiency of 49% greater than removing a non-flooded node. This indicates that the structural effects of disruptive events related to floods on urban mobility are not only statistically significant but are also substantively significant.

These findings suggest a structural overlap between flood exposure and network criticality, meaning that environmental hazards tend to impact nodes that are essential for maintaining urban mobility.

4 Conclusion and Future work

This study presented a graph-theoretical framework for assessing the vulnerability of urban transportation networks under flood-related disruptions. By modeling the road system as a spatial network and applying betweenness centrality and global efficiency analysis, it was possible to identify structurally critical components and evaluate how localized disruptions propagate through the system.

The results demonstrate that flood-affected nodes exhibit significantly higher betweenness centrality and vulnerability than randomly sampled non-flood nodes, with flood-affected nodes showing a median vulnerability approximately 49% greater than non-flood nodes, corresponding to a 0.24% versus 0.16% relative reduction in global network efficiency upon removal. These findings confirm a structural overlap between flood exposure and network criticality, indicating that environmental hazards disproportionately impact intersections that are essential for maintaining urban mobility. The proposed approach contributes to the identification of critical infrastructure elements that should be prioritized in urban planning and disaster risk mitigation strategies.

Future work may extend this analysis by incorporating dynamic traffic demand, probabilistic flood scenarios, and multilayer infrastructure networks. Such extensions would allow a more compre-

hensive assessment of system behavior under extreme events and further strengthen the role of network science in urban resilience studies.

Acknowledgements

This study was funded by the Brazilian Federal Agency for Support and Evaluation of Graduate Education (CAPES) - Finance Code 001 (E.I.B. and G.G.S.); by the São Paulo Research Foundation (FAPESP), grant no. 2024/02748-7 (E.V.E.S.); and by the Brazilian National Council for Scientific and Technological Development (CNPq), grants no. 446053/2023-6 and no. 305205/2025-0 (L.B.L.S.).

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